

Day 11 15/12/2025

Data Structures

Data structure is a way of organizing, storing & managing data so that it can be accessed and modified efficiently.

Common data structure in python are:-

1. List
2. Tuple
3. Set
4. Dictionary

I. List

A list is ordered, mutable (changeable) collection of data

Key features

- Allow duplicate values
- Elements can be of different data types
Ex:- ['python', 2, 3, 35.5]
- Index start from '0'

Ex:- l1 = [1, 2, 3, 4, 5]

l2 = [4, 5, 6, 7, 8]

l3 = []

for i in range(len(l1)):

l3.append(l1[i] + l2[i])

print(l3)

Slicing in list

* It is used to extract a part (sublist) from a list

list_name [Start : Stop : step]

Ex:- l = [10, 20, 30, 40, 50, 60, 70]
l[1:5]
op = [20, 30, 40, 50]

2. Tuples

A tuple is an ordered, immutable collection of items

Key features

- * faster than lists
- * Cannot be modified after creation
- * Allows duplicates.

Ex:- t₁ = (1, 2, 3, 4, 5)
t₂ = (1, 2, 3)
t₃ = t₁ + t₂
t₃
op = (1, 2, 3, 4, 5, 1, 2, 3)

Ex 2 :- temp1 = list(t₃)
temp1[1] = "Kiran"
t₃ = tuple(temp1)
t₃
op = (1, 'Kiran', 3, 4, 5, 1, 2, 3)

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3. Sets

It is a multivalued and heterogeneous data collection. It is mutable, unordered, and it won't allow duplicates.

*It is represented by {} or set()

unordered

Ex:- $S_1 = \{12, "riya", 35.5, True, 12, 12+3j, "latha"\}$

$op = \{(12+3j), 12, 35.5, True, 'latha', 'riya'\}$

To Add

$S_1.add(3)$

S_1

$op = \{(12+3j), 12, 3, 35.5, True, 'latha', 'riya'\}$

To Remove

$S_1.remove(35.5)$

S_1

$op = \{(12+3j), 12, 3, True, 'latha', 'riya'\}$

① Union

$$S_1 = \{1, 2, 3, 4, 5\}$$

$$S_2 = \{4, 5, 6, 7, 8\}$$

$$S_1 \cup S_2$$

$$op = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

② Difference

$$S_1 = \{1, 2, 3, 4, 5\}$$

$$S_2 = \{4, 5, 6, 7, 8\}$$

$$S_1 - S_2$$

$$op = \{1, 2, 3\}$$

④ Intersection

$$S_1 = \{1, 2, 3, 4, 5\}$$

$$S_2 = \{4, 5, 6, 7, 8\}$$

$$S_1 \cap S_2$$

$$op = \{4, 5\}$$

③ Symmetric difference

$$S_1 = \{1, 2, 3, 4, 5\}$$

$$S_2 = \{4, 5, 6, 7, 8\}$$

$$S_1 \oplus S_2$$

$$op = \{1, 2, 3, 6, 7, 8\}$$

H. Dictionary

It is multivalued heterogeneous data collector & it collects the data in {key: value} here key is immutable and value is mutable

- * It is ordered and only values will allow duplicates
- * It is represented by {key: values} or dict()
- * whenever we need to update @ remove we need to do with key.

Ex:- dict = {'name': 'Chandana.R',
'age': 32,
'degree': 'B.Tech'}

dict

-> To update age

Ex:- dict['age'] = 23
dict

op = {'name': 'Chandana.R',
'age': 23,
'degree': 'B.Tech'}

5. String :

It's a sequence of data type and immutable in nature and it can hold all type of data.

Since it's ordered let check

`s1 = "besant123@tch!@#gmail.com"`

`s1`

`op = "besant123@tch!@#gmail.com"`